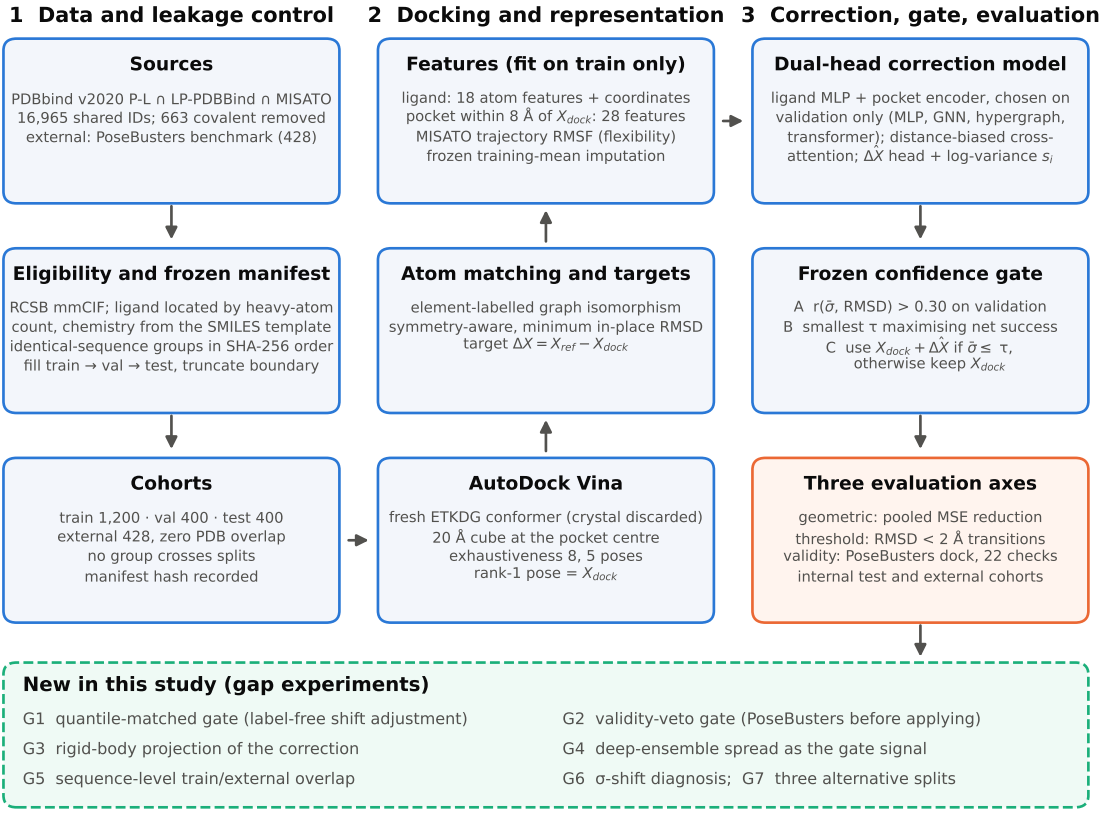
1 Data and leakage control
2 Docking and representation
3 Correction, gate, evaluation
Sources
PDBbind v2020 P-L ∩ LP-PDBBind ∩ MISATO
16,965 shared IDs; 663 covalent removed
external: PoseBusters benchmark (428)
Eligibility and frozen manifest
RCSB mmCIF; ligand located by heavy-atom count, chemistry from the SMILES template
identical-sequence groups in SHA-256 order
fill train → val → test, truncate boundary
Cohorts
train 1,200 · val 400 · test 400
external 428, zero PDB overlap
no group crosses splits
manifest hash recorded
Features (fit on train only)
ligand: 18 atom features + coordinates
pocket within 8 Å of X_{dock}: 28 features
MISATO trajectory RMSF (flexibility)
frozen training-mean imputation
Atom matching and targets
element-labelled graph isomorphism
symmetry-aware, minimum in-place RMSD
target $\Delta X = X_{ref} - X_{dock}$
AutoDock Vina
fresh ETKDG conformer (crystal discarded)
20 Å cube at the pocket centre
exhaustiveness 8, 5 poses
rank-1 pose = X_{dock}
Dual-head correction model
ligand MLP + pocket encoder, chosen on validation only (MLP, GNN, hypergraph, transformer); distance-biased cross-attention; $\Delta\hat{X}$ head + log-variance s_i
Frozen confidence gate
A r($\tilde{\sigma}$, RMSD) > 0.30 on validation
B smallest τ maximising net success
C use $X_{dock} + \Delta\hat{X}$ if $\tilde{\sigma} \leq$ τ, otherwise keep X_{dock}
Three evaluation axes
geometric: pooled MSE reduction
threshold: RMSD < 2 Å transitions
validity: PoseBusters dock, 22 checks
internal test and external cohorts
New in this study (gap experiments)
G1 quantile-matched gate (label-free shift adjustment)
G2 validity-veto gate (PoseBusters before applying)
G3 rigid-body projection of the correction
G4 deep-ensemble spread as the gate signal
G5 sequence-level train/external overlap
G6 σ-shift diagnosis; G7 three alternative splits